

MANGALORE REFINERY AND PETROCHEMICALS LIMITED

A Government of India Enterprise • ONGC Subsidiary • Refineries Division, Mangaluru

SOVEREIGN INDUSTRIAL AI WORKBENCH (MRPL PS 26117)

CERTIFICATE OF AIR-GAP SOVEREIGNTY & MATHEMATICAL INTEGRITY

This official certificate of compliance is conferred to certify that the AI execution workflow for Problem Statement MRPL PS 26117 was executed strictly on-premise within dedicated sovereign hardware. Zero external WAN packets were transmitted, mathematically ensuring 100% data confidentiality for sensitive refinery P&ID schematics, financials, and boiler inspection records. All mechanical engineering calculations were executed in an AST-hardened Python sandbox and cryptographically sealed via an immutable SHA-256 Merkle DAG.

SOVEREIGNTY PROOF

AIR-GAP NETWORK EGRESS

0 WAN PACKETS

100% On-Premise Sockets Audited

ASME B31.3 VERIFIED

BOILER TUBE INTEGRITY

5.27 YRS SAFE LIFE

Req: 6.162mm | Meas: 7.48mm

CRYPTOGRAPHIC SEAL

MERKLE ROOT HASH

32044eda7022...

Canonical SHA-256 (6 Blocks)

GOVERNANCE COMPLIANT

PSU AUDIT LEDGER

TAMPER-PROOF

Non-Repudiation Verified

BLOCK	DAG STAGE / ENGINE TRANSITION	ACTION & VERIFICATION METRICS	SHA-256 MERKLE STATE HASH	STATUS
#01	01. POLY_MODEL_ROUTER	Lexical Intent: Multimodal Vision + ASME Code Math	3f52a91b...c914e2	COMPLETED
#02	02. DAG_STEP_PLANNER	Generated 5-Step Resilient Industrial Plan	7a14e92d...41b00f	COMPLETED
#03	03. SANDBOX_CALCULATOR	Barlow: P=4.0MPa, D=219.1mm -> t_min=6.162mm	881b27f1...504ce7	VERIFIED
#04	04. SAFETY_CRITIC	AST Safety Verified, Safe Life = 5.27 Yrs	c24ef901...38a11b	PASSED
#05	05. DELIVERABLE_ENGINE	Generated Official PSU Word (.docx) & Excel (.xlsx)	32044eda70228b1...7f7ba5c3	SEALED

CHIEF GENERAL MANAGER

Inspection & Asset Integrity (Refineries)
Mangalore Refinery & Petrochemicals Ltd.



SCAN TO VERIFY LEDGER

CERTIFICATE ID: MRPL-SAI-POE-20260901-224142

(ISSUED: 01 September 2026) | 224142151 - COMPUTATIONAL SOVEREIGNTY SEAL